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Design and analysis of personalized serious games for information literacy: catering to introverted and extraverted individuals through game elements

Phimphakan Thongthip[1], Kannikar Intawong[2], Suepphong Chernbumroong[3], Perasuk Worragin[3], Yootthapong Tongpaeng[4] & Kitti Puritat[1]✉

Personalized learning has become increasingly prevalent in computer-based education. Nevertheless, there remains a scarcity of studies addressing personalized serious games. This paper delves into a study on a personalized serious game that utilizes suitable game elements tailored to students' personality traits, focusing on the dimensions of introversion and extraversion. To evaluate the impact of personalized serious games, a comparative experiment was conducted. The study involved both a Control group (Non-personalized game-based approach) and an Experimental group (Personalized game-based approach). Participants were assessed using pre-post tests measuring knowledge acquisition and retention in information literacy for source evaluation, as well as intrinsic motivation measured through the IMI questionnaire. The findings indicate that while personalized serious games can enhance intrinsic motivation, particularly in terms of perceived competence and effort, they did not significantly impact knowledge acquisition and retention. User behavior data statistics revealed a substantial 37% improvement in engagement, measured by both average and total playtime, especially noticeable among introverted participants. However, this personalized approach was found to be less effective for extraverted participants.

[1] Department of Library and Information Science, Faculty of Humanities, Chiang Mai University, Chiang Mai, Thailand. [2] Faculty of Public Health, Chiang Mai University, Chiang Mai, Thailand. [3] College of Arts, Media and Technology, Chiang Mai University, Chiang Mai, Thailand. [4] School of Information Technology, Mae Fah Luang University, Chiang Rai, Thailand. ✉email: kitti.p@cmu.ac.th

Introduction

People have the unprecedented ability to communicate with one another effortlessly in the contemporary era, which is characterized by the digital age, which adopts and integrates digital technologies, such as the internet and computers, facilitated by abounding amounts of information and knowledge sources. With the emergence of technological advancements, an abundance of social networking platforms has arisen, serving as the favored sources of news consumption for a large number of people. According to a study conducted by Agarwal and Dixit (2020), social media platforms have become increasingly prevalent as primary sources of news consumption. The study reveals that a significant proportion of individuals rely on social media platforms for news, with specific statistics indicating that nearly 450 million of the surveyed individuals will consider social media platforms as their primary source of daily news consumption by 2020 (Agarwal & Dixit, 2020). This expansion, which occurred at a rate of approximately one million users per month per country, resulted in a total of 4.1 billion active social media users globally (Kemp, 2019). Furthermore, Velichety and Shrivastava (2022) conducted a research study to better understand the dependency on social media platforms and search engines for news consumption. The study discovered that approximately 65% of those polled cited social media platforms and search engines as their major sources of daily news consumption. These results underline the importance of social media platforms and search engines in altering people's news-consuming habits in modern society. This immense rise, together with the current user base, demonstrates the widespread dependence on social media platforms to obtain news and information. Despite their extensive use and acknowledgment, social media platforms confront a variety of challenges, including the prevalence of online misinformation, sometimes known as fake news. Hern (2017), Allcott and Gentzkow (2017) highlight the concerning growth of online misinformation on social media platforms, attributing it to the lack of editorial oversight and the platform's inherent design, which allows for easy transmission to large audiences, spreading false information to many readers. Hern's results highlight the urgent necessity of dealing with the issue of misinformation on social media platforms to protect users from misleading information. While this issue has persisted since the inception of mass communication, the ascent of social media platforms has significantly exacerbated it. The acquisition of information literacy skills has become a significant educational endeavor in order to reduce the risk of exposure to falsehoods and protect people from falling victim to false information stories. These skills, which are an important aspect of 21st-century competence, are extensively imparted to students. Recognizing the issues posed by misinformation, developing information literacy abilities has become an important educational activity. Aydin (2021) emphasizes the need for information literacy abilities in countering misinformation and preventing people from being victims of misleading information stories The learning of these abilities provides individuals with the capacity to critically analyze information and distinguish between reliable sources and misinformation. Furthermore, educational institutions are increasingly using e-learning pedagogical techniques to teach students. Nonetheless, effective motivation and engagement of students in the usage of these digital learning platforms remain a difficulty. Poondej, Lerdpornkulrat (2019) emphasize the problems that educational institutions encounter while engaging students in e-learning platforms, underscoring the significance of resolving these issues to improve the success of digital learning efforts. Serious games have emerged as potent tools for promoting lifelong learning and assisting individuals in staying abreast of evolving trends. The current interdisciplinary applications substantiate the significance of this emerging paradigm in the context of learning (Chen et al., 2019). Furthermore, in order to render a game adaptable and enjoyable for diverse player profiles, the incorporation of game personalization has been employed as a means to augment player engagement and attentiveness.

This research investigates the impact of a personalized serious game designed to enhance and instruct information literacy skills among students, specifically focusing on the differences between introverted and extroverted individuals. The study aims to incorporate suitable gaming elements as tools necessary to effectively combat the dissemination of fake news across various mass media platforms.

Literature review

Fake news and information literacy. In the delineation of false or deceptive information in news, extensive investigations by researchers and journalists have led to the dissemination of their findings. Numerous initiatives have been undertaken to discern and differentiate misleading information. For example, Wang (2020) defined fake news as "reports disseminating erroneous or inaccurate information while failing to acknowledge the inaccuracies within". Park et al. (2020) established a comprehensive framework for categorizing fake news based on the intent to deceive and inflict harm, comprising four distinct categories: non-information, disinformation, misinformation, and mal-information (Park et al., 2020). Non-information involves substantial intent to deceive with minimal intent to inflict harm, while disinformation denotes significant intent for both deception and harm. Misinformation indicates minimal intent for both deception and harm, while mal-information involves limited intent to deceive with a pronounced intent to cause harm (Park et al., 2020). Despite the presence of four separate types of fake news: non-information, disinformation, misinformation, and mal-information, this study centers solely on misinformation. The definition of misinformation has been presented by several researchers. Pennycook and Rand (2019) define misinformation as "inaccurate or misleading facts that are disseminated with the intention to mislead." Lewandowsky, Ecker, and Cook (2017) define misinformation as "information deemed inaccurate or misleading." In their study, Vosoughi et al. (2018) define misinformation as "bogus data that is conveyed without considering a motive to mislead." Tandoc (2021) argued that fake news represents a distinct category of deliberately crafted falsehoods mimicking legitimate news content. Consequently, individuals must possess essential skills for information retrieval and utilization due to the prevalent use of digital technology and the interconnectedness of the Internet in today's information-centric society.

To combat fake news, information literacy can serve as an educational tool. Information literacy, defined differently by various authors, showcases an individual's ability to determine their information requirements, efficiently search for information, critically assess procured information, and apply it to achieve their objectives (Guo & Huang, 2020). It involves defining information needs, managing information sources effectively, and proficiently using digital technology for sourcing, analyzing, and synthesizing information resources (Pinto et al., 2020). Information literacy encompasses critically evaluating the credibility of resources, utilizing appropriate citation methodologies, adhering to legal and ethical considerations, and formulating research inquiries precisely and efficiently (Reddy, et al., 2020). However, the Internet poses risks, especially for those lacking information literacy skills. In the 21st century, education emphasizes lifelong learning skills, where information literacy forms the foundation for adapting to an ever-evolving information landscape (White (2019)). White's findings support that individuals equipped with information literacy and lifelong learning skills are better prepared to face the multifaceted challenges of the 21st century (White (2019)). Therefore, information literacy has evolved into an indispensable skill set for contemporary survival and proficiency (Banik & Kumar, 2019).

Serious games for education. Serious video games have increased in popularity being used in educational contexts because of their ability to improve educational experiences and performance. Such games are created with particular educational objectives in mind, seeking to involve students in engaging and immersive experiences that enhance knowledge acquisition, improve skills, and critical thinking. In contrast to traditional educational techniques, serious games provide an innovative and interactive atmosphere for learning that fosters active involvement and engagement (Chen & Michael, 2005). Landers (2014) differentiates serious games from those aimed purely at amusement by focusing on promoting learning in various forms. These games attempt to improve players' understanding of various subjects while also helping to change their attitudes and views on such subjects. Serious games create informative material during gameplay, usually in the form of raw user data that represents the player's actions inside the game. This information reveals the learner's techniques for finding, acquiring, and digesting knowledge. Vidakis et al. (2019) demonstrate how distinct serious games gather varying datasets, including queries, user interactions, and task completions, which are used to compute educational progression results. Connolly et al. (2012) underline various well-established benefits of using serious games in education, such as active learner involvement, facilitated and seamless learning experiences, timely feedback, and a significant increase in student motivation and engagement. Romero et al. (2015) go on to assert that well-designed serious games have proven to be very effective instructional tools, boosting students' learning capacity by encouraging participation and generating critical thinking throughout the educational process. Furthermore, Ketamo et al. (2018) highlight how educational video games offer the customization of learning material and manner of delivery in order to accommodate customized learning experiences. This personalizing component improves the flexibility of educational interventions, making them more successful at fulfilling learners' different requirements and preferences.

Personalized educational games. Personalized educational games have developed as a potential strategy for meeting the different educational requirements and preferences of individual students. These games are intended to customize instructional material and learning experiences depending on each student's unique features, talents, and interests, thereby increasing engagement and improving successful learning results. Personalized learning systems possess the capacity to adapt the presented educational content in real-time, tailoring it to the specific preferences, abilities, and existing knowledge of individual learners. Furthermore, these systems have the capability to individualize instructional methods to optimize the learner's performance. Consequently, they offer solutions to several challenges commonly encountered in traditional learning environments, including resource constraints, learner engagement, and the diversity of learners' knowledge and preferred learning modalities (Tlili et al., 2019). Hence, to accommodate the distinctive needs of learners, numerous researchers and educators have contemplated the utilization of personalized learning systems as an alternative to the conventional one-size-fits-all approach (West, 2011). Hwang and Tsai (2011) presented the "Personalized Learning Environment" (PLE) framework, which is an example of customized games for learning. The PLE framework takes into account a variety of student characteristics, including their cognitive style, approach to learning, and previous knowledge, to personalize educational material and tasks according to learners' requirements. PLEs provide specific educational experiences that increase the effectiveness of learning by utilizing unique learning methods. Additionally, devised an educational system known as LearnFit, which employs the Myers Briggs Type Indicator (MBTI) to assess the learner's personality. Subsequently, it tailors a personalized learning approach for each individual based on their personality traits. Nevertheless, numerous personality models featuring diverse personality dimensions have been elucidated in the academic literature (Tlili et al., 2019). Among these dimensions, the introvert-extravert dimension is a prominent element, commonly observed in well-known personality models like the Big Five Factor and Myers-Briggs (Tlili et al., 2019). In addition to personality, learning styles, cognitive capacities, and past knowledge are important components in customized learning systems (Newman et al., 1995). Furthermore, technological advancements such as artificial intelligence and machine learning are improving the potential of personalized educational platforms, allowing for greater accuracy and effective adjustment to learners' demands (VanLehn, 2006).

Aspects of introverted and extraverted personality types. Personality traits influence people's habits of study and attitudes toward education. According to study findings, traits of personality, including extraversion, introversion, openness to experience, conscientiousness, agreeableness, and neuroticism, impact how people perform learning activities and engage in educational contexts (McCrae & Costa, 1987). Extraverts, for example, may favor collaboration environments for learning and prefer social engagement in the course of their studies, whereas introverts may favor isolated and independent study (McCrae & Costa, 1987). Mount et al. (2005) characterize personality traits as enduring psychological attributes that delineate individuals' conduct and cognitive approach. Extensive research in the literature has demonstrated that personality exerts influence on learners across multiple dimensions, including their perception of educational systems (Cohen & Baruth, 2017), preferences for game genres and design elements in educational environments (Denden et al., 2018; Schimmenti et al., 2017), and their learning performance (Anderson et al., 2018). Moreover, extraversion and introversion preferences contribute to people's perceptions of education, affecting their perspectives, drive, and participation in educational activities (Poropat, 2009). Extraverted people may see school as a chance for social engagement, connections, and personal development, whereas introverted people may perceive it as a method of intellectual study, discovering themselves, and contemplation (Poropat, 2009). Chen et al. (2016) discovered that variances in learners' personalities can significantly influence their learning behaviors within the context of Massive Open Online Courses (MOOCs). As mentioned in the previous section, this study focuses on the introvert-extravert dimension. The designations "introversion" and "extraversion" trace their origins to the 1920s and are credited to the work of the psychologists Jung and Baynes (1923). The concept of introversion pertains to individuals who derive their vitality from internal sources such as thoughts, imagery, and contemplation. Conversely, extraversion refers to individuals who are action-oriented and derive their vitality from the external world (Tlili et al., 2019). Costa, McCrae (2008) have delineated six facets corresponding to each trait within the Big Five Factor model, including extraversion,

Table 1 A summary of the differences between extraversion and introversion traits.

Characteristics	Extraversion	Introversion
Social interactions	Enjoys socializing with a large group of friends (McCrae & Costa, 1989)	Prefers isolation and close connections (Asendorpf, 2000)
Behavioral tendencies	Outgoing, assertive, and sociable (Costa, McCrae (2008))	Reserved, self-reflective, and thoughtful (Eysenck (1977))
Emotional expression	Expressive, external expressions of emotion (Watson & Clark, 1997)	Reserved, lowered expression of emotion (Carver & White, 1994)
Cognitive styles	Action-oriented; seeks stimulation from outside sources (DeYoung, 2015)	Reflective, profound thought, continuous focus (Costa & McCrae 1980)

neuroticism, agreeableness, conscientiousness, and openness to experience. In the context of extraversion, the six facets, as defined by Costa, McCrae (2008), are outlined as follows:

1. Warmth: These individuals are characterized by their amiable nature, demonstrating friendliness and affection towards others.
2. Excitement seeking: They exhibit a propensity for experiencing boredom quickly and actively seek excitement and stimulation.
3. Activity: They are marked by their high energy levels, liveliness, and preference for physical movement.
4. Assertiveness: These individuals exude self-assurance and often take on leadership roles within their social groups.
5. Gregariousness: They have an aversion to solitude and generally favor the company of others.
6. Positive emotions: They frequently experience a profusion of positive emotional states.

Introversion is characterized by being thoughtful, less inclined towards socializing, often reserved in speech, and feeling self-conscious when engaging in social interactions. Introverts typically enjoy smaller gatherings or one-on-one interactions, preferring to take their time getting acquainted with someone new. However, it is important to recognize that introverts simply express their sociality in unique ways, valuing personal connections, privacy, and tranquility (Paradilla et al., 2021). Introverts typically enjoy moments of solitude, favoring self-reflection over outward expression, prioritizing depth in their interactions, and often showing less emotional demonstration. They tend to confide personal information in a small circle of trusted individuals. Writing may be more appealing to introverts than verbal communication, and they may occasionally experience social fatigue, seeking consolation in solitude to recharge their energy (Dossey, 2016). The book by Grimes et al. (2011) "Four Meanings of Introversion: Social, Thinking, Anxious, and Inhibited Introversion" studies many aspects of introversion. Following is a description of each type:

1. "Social Introversion" is mainly concerned with interaction with others. People with social introversion may favor isolation or small groups over significant social gatherings. They prefer to recharge by enjoying themselves on their own or with a small group of companions, as opposed to huge groups.
2. "Thinking introversion" refers to how people organize knowledge for themselves. These people can devote a lot of time to meditating on their ideas and thoughts, valuing reflective thinking over external expression. They can succeed at tasks that involve profound conceptualization or creative thinking.
3. "Anxious Introversion" type is characterized by an increased response to social interactions as well as an inclination for sensations of unease in situations that involve others. People with such a type of introversion can become self-conscious or uneasy around other people, which causes them to refrain from some social gatherings or encounters.
4. "Inhibited Introversion" indicates a guarded or wary attitude toward interacting with other people. Those with this form of introversion can be cautious when interacting with other people or conveying themselves completely. They can choose to watch instead of taking part in social events, usually spending their time relaxing before utterly participating.

Nevertheless, according to Blevins et al. (2022), introversion can occasionally be interpreted incorrectly as being a complete contrast to extraversion. The misunderstanding associates introversion with unfavorable valenced features, such as awkwardness in social situations as well as low social self-esteem. This misperception comes from an inclination to perceive extraversion and introversion as opposite in characteristics, with introverts perceived as having no ability to interact with others or confidence when contrasted to extroverts. Nevertheless, this simplification dismisses the richness and variety that come with every character type. Table 1 provides a summary of the distinctions between extraversion and introversion traits, focusing on empirical data and theoretical frameworks.

Personality traits are notably linked to the learning styles of individuals. Extraversion and introversion are two essential personality types that have a major effect on people's ways of learning and attitudes toward education. Extraversion is defined by socialization, assertiveness, and a desire for outward stimulation, whereas introversion is distinguished by contemplation, reserve, and an inclination for the inside thought (Costa, McCrae (2008)). Introvert learners are characterized as reflective thinkers, gravitating towards more contemplative learning approaches; in contrast, extravert learners exhibit a preference for active and interactive learning methodologies (Felder & Silverman, 1988). Individuals with a high level of extraversion excel in interactive and social educational environments, enjoying collaborative tasks, and discussions in groups. According to research, extraverts tend to favor active learning techniques that involve group assignments and hands-on tasks, which encourage social connection and participation (Borg & Shapiro, 1996). Furthermore, extraverts can try out extracurricular events and leadership positions to enhance their overall school experience and personal growth (Borg & Shapiro, 1996). Introverts, on the other hand, may favor isolated and independent study, as well as serene and concentrated situations that enable deep thought and concentration (McCrae & Costa, 1987). Introverts may succeed in self-paced educational settings where they can investigate issues at their own speed and dive deeply into complicated topics (McCrae & Costa, 1987). Furthermore, introverts may favor written communications and written reflection projects that allow them to communicate their ideas and opinions. Moreover, several research investigations have underscored the impact of personality on the

Table 2 Summary of the effects of game elements on introverts and extraverts.

Game element	Objective	Extravert	Introvert	Implications for game design
Badge	A visual depiction or icons of the user's accomplishments (Denden et al., 2017)	Positive	Neutral	Extraverts exhibited a greater favor for badges (Codish, Ravid (2014a); Codish, Ravid (2014b))
Daily Login Reward	Reinforcements applied consistently to shape and reinforce the user's behavior, fostering the anticipation of forthcoming rewards (Fernandes, Junior (2016))	Neutral	Positive	Introverts showed a high favor for Login (Smiderle et al., 2020)
Leaderboard	Displays user ranks based on specific criteria such as points, levels, or badges, providing context to other game elements by facilitating user comparisons (Denden et al., 2017)	Positive	Negative	Extraverts displayed significant favor toward Leaderboards (Jia et al., 2016)
Personal Ranking	Users' relative placement in a list (Denden et al., 2017)	Neutral	Positive	Introverts were motivated by Ranking (Jia et al., 2016)
Progress bar	Monitoring users' advancement through the system's objectives, facilitating a visual representation of their progress over time (Denden et al., 2017)	Positive	Negative	Introverts were described not enjoying competing (Jia et al., 2016)
Points	A form of numerical feedback given to users upon performing specific actions (Denden et al., 2017)	Positive	Neutral	Extravert users were motivated to show their achievements (Jia et al., 2016)
Reward	Any reward that a user obtains in response to their actions (Denden et al., 2017)	Positive	Neutral	Reward perceived to be the most entertaining (Jia et al., 2016)
Avatar	A player's representation, which may take various forms, including virtual, physical, or even the players themselves (Ferro, 2021)	Neutral	Positive	Introverts may prefer digital means of communication and may be more inclined to personalize their avatars for self-expression (Baker et al., 2021) Customization options for introverted players to personalize their avatars, enhancing engagement and immersion.

cognitive load experienced by individuals (Gray et al., 2005). Within this framework, Eysenck (1967) posited within his personality model that introverted and extraverted individuals may exhibit distinct cognitive loads contingent upon the level of arousal present in a given learning environment. As demonstrated by Nuckcheddy (2018) through a comprehensive literature review, personality traits, particularly introverted and extraverted characteristics, can exert a discernible influence on the motivation levels of individuals within a workplace. Introverted and extraverted tendencies exhibit distinct approaches to dealing with the two motivational elements delineated in Frederick Herzberg's motivation theory, namely motivators and hygiene factors. For example, individuals with introverted tendencies tend to adeptly address hygiene factors like supervision and interpersonal relationships, while those with extraverted inclinations effectively navigate the existence of intrinsic motivators such as attainment and acknowledgement (Herzberg (2008).

Game elements for learner personalities. Learner personalities have been observed to exert influence over their inclinations towards specific game elements. For example, in the study by Jia et al. (2016), it was noted that extravert learners displayed significantly favorable attitudes toward utilizing game elements like leaderboards and progress bars, whereas their introvert counterparts did not share this preference. In other words, extraverted individuals exhibited a higher level of motivation when exposed to points, levels, and ranking elements. Moreover, researchers observed that individuals with extraverted personalities tended to derive greater enjoyment from rewards (Jia et al., 2016). Additionally, Codish, Ravid (2014a) discovered that extravert learners exhibited a greater inclination for using badges compared to introvert learners. Besides, they conducted a study using preference questionnaires to investigate how individuals with extraverted and introverted characteristics perceived gamification. It was observed that extraverts exhibited a preference for badges in this context (Codish, Ravid (2014b)). Further, participants exhibiting introverted characteristics, within both the Experimental and Control groups, demonstrated a notably increased accumulation of logins in comparison to the extraverted participants (Smiderle et al., 2020). Besides, the inclusion of an avatar element can provide a degree of anonymity to introverted individuals, facilitating their self-expression in a more carefree manner. Introverted users found that avatars offered novel avenues for adjusting their appearances to facilitate communication in various contexts (Baker et al., 2021). Table 2 demonstrates the summary of the effects of game elements on introverts and extraverts.

Purpose of study and research questions

Regarding the creation of an effective serious game, numerous research studies have highlighted the potential for designing personalized games to enhance various aspects of learning, including knowledge acquisition (Pakinee & Puritat, 2021), motivation (Harteveld & Sutherland, 2017), and technology acceptance (Svendsen et al., 2013; Maican et al., 2019). However, to the best of our current knowledge, no study has reported the implementation of personalized serious games that consider learners' personalities by implementing game elements to impact learner outcomes in knowledge acquisition, motivation, knowledge retention, and learner behavior. Therefore, this study contributes to the existing literature by developing a personalized serious game for learners focusing on information literacy and fake news. It is grounded in the Big Five theory, with specific emphasis on extraversion and introversion. The study's aim is to explore the effects of personalization on motivation, knowledge acquisition, knowledge retention, and user behavior within the context of our online serious game. To this end, the study addresses the following research questions:

(RQ1) Is there a significant difference in enhancing knowledge acquisition between Personalized and Non-personalized serious games?

(RQ2) Is there a significant difference in enhancing knowledge retention between Personalized and Non-personalized serious games?

(RQ3) Is there a significant difference in enhancing learning motivation between Personalized and Non-personalized serious games?

(RQ4) What is the effect on engagement for extraversion and introversion personality traits in Personalized and Non-personalized serious games?

Addressing these research questions carries substantial advantages for the field of library and information science, as well as the broader educational and media literacy domains. Through an exploration of how personalization within serious games impacts learners' motivation, knowledge acquisition, and knowledge retention, this study can offer invaluable insights that extend to the development of effective educational resources. These findings can be of immense practical value to libraries and educators, serving as a guideline for librarians and researchers aiming to design personality-centered serious games. This research can facilitate the creation of tailored educational tools that enhance information literacy, equipping librarians and educators with powerful resources to instruct and guide users effectively in navigating the complexities of the information landscape.

Fig. 1 NPCs provide the knowledge of information literacy.

Fig. 2 How players can learn about the impact of misinformation on the city.

Game design and implementation

Libraries and Library Information Science (LIS) professionals play a pivotal role in guiding user communities to access and critically evaluate accurate information (Agosto, 2018; De Paor & Heravi, 2020). They are also responsible for developing tools that enhance media and digital literacy competencies to combat the proliferation of fake news. In line with this mission, the Thailand's Safe and Creative Media Development Fund has collaborated with the Department of Library and Information Science at Chiang Mai University to develop an online serious game aimed at enhancing information literacy and addressing the issue of misinformation among students in higher education and the wider community.

In this study, we utilize an online serious game as an educational tool for teaching information literacy within the university curriculum. Information Literacy is a general subject open to all students, and it focuses on equipping them with the essential skills to critically assess information, particularly in today's digital age characterized by the proliferation of false or misleading information. This curriculum includes teaching source evaluation, fact-checking, and using methods like the Currency, Relevance, Authority, Accuracy, and Purpose (CRAAP) test (Liu, 2021) to evaluate sources for their credibility and reliability.

The game design concept centers on a 2D platformer side-scrolling Role-Playing Game (RPG). In this design, learners take on the role of librarians responsible for managing the news content to be disseminated by the National Library of Thailand to the broader community. The primary task for players is to assess and verify news sources gathered from various social media platforms. Their objective is to distinguish between accurate information and misinformation. Subsequently, players investigate the impact of the news through city observations and by listening to residents. The game's activities encompass key steps: engagement with citizens, observation of the city, and evaluation of news sources. Each of these steps will be described in more detail below.

Engagement with citizens. The initial game activity is intended to facilitate learning about the impact of misinformation by engaging with the city's residents. Players can control their character, walking and communicating with citizens from their homes, progressing through various areas including rural zones, hospitals, the city center, and business communities before reaching the workplace. The game's mechanics are designed to help players understand people's perspectives on news and its effects. Additionally, during this game activity, "men in black" non-player characters (NPCs) have been incorporated to impart information literacy concepts and the CRAAP test for source verification to players along the way through pop-up messages, as depicted in Fig. 1.

Observation of the city. This key game activity aims to simulate the effects of fake news on a broader community and society. Similar to the previous game activity, players can control their character to walk through the city and observe the impact of fake news. The game design includes five categories of news: education, economy, social matters, health, and religion and belief. This game design is a crucial concept for learners to understand and become aware of the impact of misinformation. It illustrates how a single piece of misinformation, when disseminated to the wider community, can rapidly have a serious impact, as shown in Fig. 2.

Evaluation of news sources. This activity constitutes the most critical game design element for learners to enhance their news source evaluation skills. The game utilizes a quiz-based format where users are assigned the task of assessing six news items on their table daily, spanning various categories. Players must thoughtfully determine which of these news items are fake and which are authentic before delivering them to the city's residents. The primary objective of this learning activity is to enhance skills in evaluating news using the CRAAP test. The game mechanics provide real-time feedback to players in the form of scoring points. The gameplay for this activity is depicted in Fig. 3.

In summary, the game is structured around a seven-day time limit for play and involves verifying approximately forty news items to determine their authenticity. Each round of play takes approximately 2–3 hours to complete. The game design is intentional, aiming to foster engagement and replayability (Adellin et al., 2019) for players by offering numerous alternative scenarios of fake news that can be influenced by the player's choices. This design approach enhances the game's value for continuous play. For instance, a player might be aware that COVID-19 cannot spread through the air, but they may choose to characterize it as real news to observe the impact on society and individuals. We used the Unity game engine to develop this game, and it is available on two platforms: PC-based and online web browsers. Additionally, for research purposes, two versions of the game were developed to cater to introverted and extraverted players by incorporating suitable game elements.

Fig. 3 The player's task of evaluating the authenticity of news articles to distinguish between real and fake information.

Game design elements for introverts. Introverted learners exhibit a preference for thoughtful and reflective approaches to education (Felder & Silverman, 1988) often finding more satisfaction in solitary tasks than in group settings. Consequently, in a gaming context, they might not favor making their activities visible to others (Jia et al., 2016). For instance, the act of logging in can significantly motivate introverted learners as it's a solitary task (Smiderle et al., 2020). They may choose to engage in daily login tasks individually, avoiding group activities or competitions.

Due to introverted learners' tendency toward lower self-confidence, a gaming system could integrate game elements or rewards, particularly an individual ranking system, designed to boost their confidence within the gaming environment (Tlili et al. (2019)). Incorporating an avatar element provides introverted individuals with a sense of anonymity, enabling them to express themselves more openly. Avatars offer a unique opportunity for introverted individuals to modify their visual appearance, facilitating communication in various situations (Baker et al., 2021). In our experiment, which specifically targeted the introvert group, we developed a game design tailored for introverted individuals. This design integrated three core game elements—Avatar, Daily Reward, and Ranking System—into the comprehensive framework of a serious game. Through this customized version, we conducted an assessment of the effects of a personalized serious game, explicitly designed for introverted players (Fig. 4).

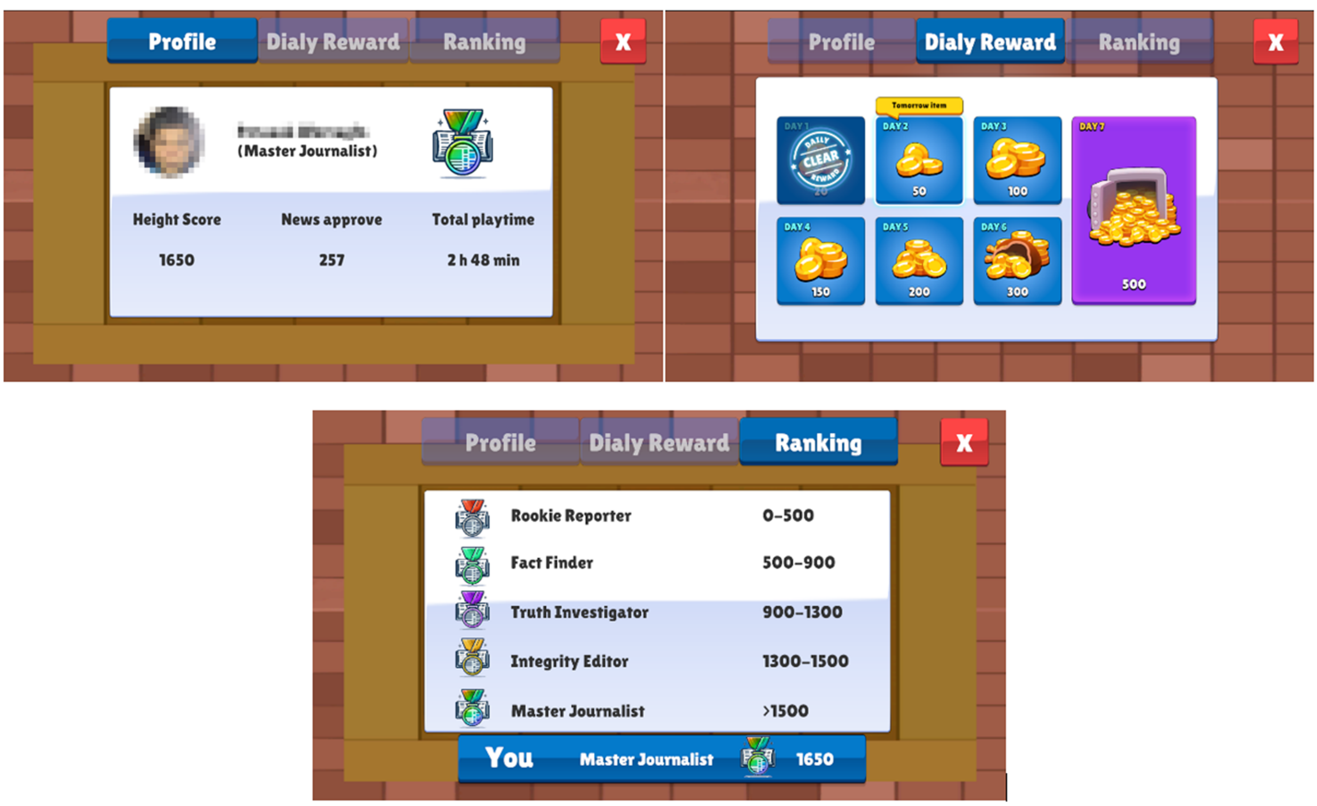

Fig. 4 The design of game elements applied to the introvert-focused serious game version.

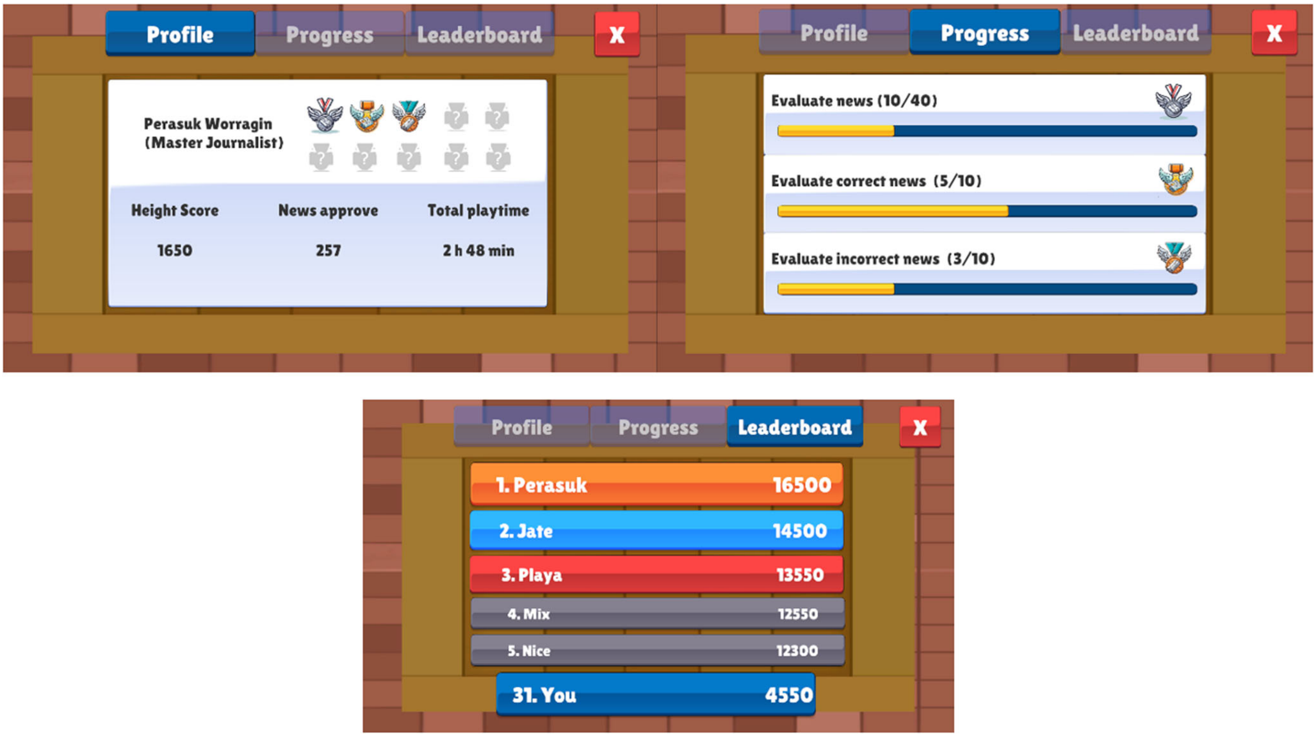

Fig. 5 The design of game elements applied to the extravert-focused serious game version.

Game design elements for extraverts. Extraversion characterizes individuals who derive their energy from external stimuli and possess an action-oriented nature (Tlili et al. (2019)). Extravert learners favor dynamic and participatory learning approaches, often showing a preference for socially competitive activities and a tendency to showcase their achievements (Jia et al., 2016). They are inclined to actively engage in social systems, particularly when a larger audience is involved (Nov & Arazy, 2013).

Enjoying the spotlight and social prominence, extraverts are drawn to elements such as the Progress system, Levels, and Leaderboards, perceiving Rewards as particularly enjoyable. To enhance prolonged engagement among extraverted users, the inclusion of Leaderboards appears to be an effective choice. Leaderboards offer a platform for users to interact within a dynamically evolving social group, catering to and adjusting to the actions of extraverted users (Jia et al., 2016). In our study focusing on the extravert group, we created an Extravert version with specifically tailored game elements—Badge, Progress, and Leaderboard—for extraverts. Notably, the Badge and Progress elements allow other players to view and compare achievements on the leaderboard, similarly to the application of introvert game elements, as depicted in Fig. 5.

Research methodology

This study employs a quasi-experimental design to investigate the effectiveness of "How to Spot Fake News," a serious game that offers two versions. The first version is designed for a Non-personalized group, while the second version is tailored to accommodate the learning preferences of introvert and extravert participants. The primary goal is to assess how these versions contribute to enhancing participants' information literacy and their ability to discern fake news. Quasi-experimental designs are particularly well-suited for educational research, as they allow for the evaluation of interventions in real-world settings while controlling for various factors that might impact the outcomes (Maciejewski, 2020).

Participants. We recruited undergraduate students enrolled in Information Literacy and Information Presentation courses at Chiang Mai University. These courses collectively enroll approximately 1000 students each year from various academic faculties. The rationale behind selecting participants from these courses was to ensure a relatively consistent baseline level of knowledge regarding information literacy and fake news. Subsequently, we extended invitations to enrolled students to participate in our experiments. To encourage their active engagement during the online experiments, we informed them about a financial compensation of 200 baht (approximately 5 USD). This compensation aimed to acknowledge the time and effort invested by participants and incentivize them to provide high-quality data for the questionnaire while encouraging their long-term participation to facilitate data collection for knowledge retention. Through these recruitment techniques, a total of 86 students volunteered to participate in the experiment. All participants were required to complete the Big Five Inventory (BFI) questionnaire (Zhao & Seibert, 2006), a widely recognized instrument in the field of psychology for assessing personality traits. Participants were selected based on an even distribution across gender and personality traits. The weekly gameplay sessions ranged from 1 to 5 hours per week. Finally, our sample consisted of 58 participants aged between 20 and 22, with an average age of 20.91, evenly divided into 29 males and 29 females. This ensured a balanced representation of introverted and extraverted personalities, who were randomly assigned to Control and Experimental groups to maintain an equal distribution of personality types. The demographics of the participants are presented in Table 3.

Instruments

Online serious game. The primary instrument in this study is our online serious game called "How to Spot Fake News," as detailed in Section 3. This serious game was specifically designed to provide instructions within university information literacy

Table 3 Demographics of the participants.

	N	Percent
Gender		
Male	29	50.00%
Female	29	50.00%
Personality trait		
Introverted	29	50.00%
Extraverted	29	50.00%
Weekly Gameplay in Hours		
0	0	0.00%
1–5	2	3.44%
5–10	10	17.24%
10–20	28	48.27%
> 20	18	31.03%

courses, with a particular focus on the evaluation of fake news and misinformation. The game was meticulously crafted to enhance replay value, offering players multiple alternative scenarios influenced by their choices, ensuring a continuous and engaging gameplay experience. To facilitate our research, we customized the serious game into two versions. The personalized version was tailored to cater to two personality types, introverts and extraverts, by adapting various game elements, as discussed in the design section. In contrast, the non-personalized version retained the original game design, without any customization or addition of game elements. This approach allowed us to explore the impact of personalization on user experiences and outcomes.

Pre/post tests. The pre/post tests and questionnaire instruments were designed to collect quantitative data and consisted of two parts. The first part encompassed pre/post tests aimed at evaluating knowledge acquisition and retention related to information literacy and fake news, with a specific focus on assessing the comprehension of the CRAAP test for source verification in social media. These tests consisted of 30 questions and were developed by a lecturer from the Department of Library and Information Science.

To measure the impact of the serious game on both groups, with regard to motivation, we turned to the Self Determination Theory (SDT) (Ryan & Deci, 2000). This theory distinguishes between two types of motivation that originate from within an individual. Intrinsic motivation is characterized by an innate desire to engage in an activity because it is inherently enjoyable, fulfilling, or meaningful. In contrast, extrinsic motivation is rooted in external factors and incentives, involving engagement in an activity primarily for the purpose of obtaining rewards, avoiding punishments, or meeting societal expectations. Research suggests that educational games have the potential to satisfy the psychological needs for autonomy, competence, and relatedness (Przybylski et al., 2010). The satisfaction of these needs has been positively linked to the enjoyment of human computation games, which, in turn, impacts intrinsic motivation (Pe-Than et al., 2014). This enhancement of intrinsic motivation is particularly beneficial for learning, as games stimulate effort, curiosity, and interest in learning activities within the context of serious games (Chen et al., 2017; Pakinee & Puritat, 2021). Thus, the second part of the questionnaire involved the short Intrinsic Motivation Inventory (IMI) questionnaire (Vos et al., 2011; Leenaraj et al., 2023), adapted from the original version Ryan & Deci, 2000). The IMI questionnaire comprises 14 items rated on a five-point Likert scale, ranging from "strongly disagree" to "strongly agree," covering aspects of intrinsic motivation, i.e., Effort, Interest, and Perceived Competence. Both parts are included in the Appendix.

Login data statistics platform. Throughout the experiments, participants accessed the serious game via their online accounts on the webpage. The serious game, in turn, collected essential login data, game scores, login timestamps, and time duration. These data collection forms served as the foundation for the systematicrecording and organization of login events, ensuring a structured approach to quantitative data collection. Comprehensive logs were meticulously maintained, documenting user activities, including usernames, timestamps, time duration and the outcomes of login attempts. This data statistics platform provided valuable insights into user behaviors and motivations, further enriching our data collection. Subsequently, the platform played a pivotal role in processing and analyzing the accumulated data, offering a diverse array of statistical and visual tools that facilitated the extraction of insights to effectively address our research questions.

Procedure. After recruiting the eligible 58 participants, we structured the experiment into four distinct steps. All participants completed the required questionnaires online in a designated room for all steps. The flowchart illustrating our experimental procedure is presented in Fig. 6. A comprehensive description of each of the four steps is provided below.

Step 1: Consent form, pre-test, and pre-IMI questionnaire. In the first step, participants were introduced to the research study and were informed about its purpose. They were then asked to sign the consent form, granting their informed willingness to participate in the research experiments. Following this, participants completed a pre-test designed to assess their baseline knowledge of information literacy and fake news, which was relevant to the information literacy course. Additionally, participants were administered the Pre-IMI questionnaire to measure their intrinsic motivation. Subsequently, participants were instructed to attend their regular lecture class in the information literacy course, specifically focusing on evaluating fake news and misinformation. Following the class, participants were provided with login credentials for the online experiment. It' s important to note that our serious game was designed as supplementary material for students after they had enrolled in the lecture class. This step was estimated to take approximately one hour to complete.

Step 2: Experiment. In the second step, participants were divided into two groups: the Non-personalized and the Personalized game-based groups. Gender and personality traits, including introversion and extraversion, were evenly distributed among both groups, as shown in Fig. 6. Notably, participants were unaware of the group to which they had been assigned. Subsequently, participants were briefed on the duration of gameplay, which was limited to two weeks. During this period, participants were encouraged to engage with the game as much as they wished, with the requirement of completing the game at least once. However, our game design was thoughtfully constructed to maximize replay value, recognizing that the skill of evaluating news sources benefits from practice and repetition. This approach allowed us to investigate participant motivation and behavior effectively. It's important to note that we customized the login system to direct participants to the appropriate game version based on their group assignment. This step was estimated to take approximately two weeks to complete, after that the login account was expired.

Step 3: Post-test, post-IMI questionnaire, and statistical data collection. Following the completion of the serious game experiment, participants from both groups gathered in a designated room to undertake a post-test, which assessed their knowledge acquisition related to information literacy. Additionally, participants filled

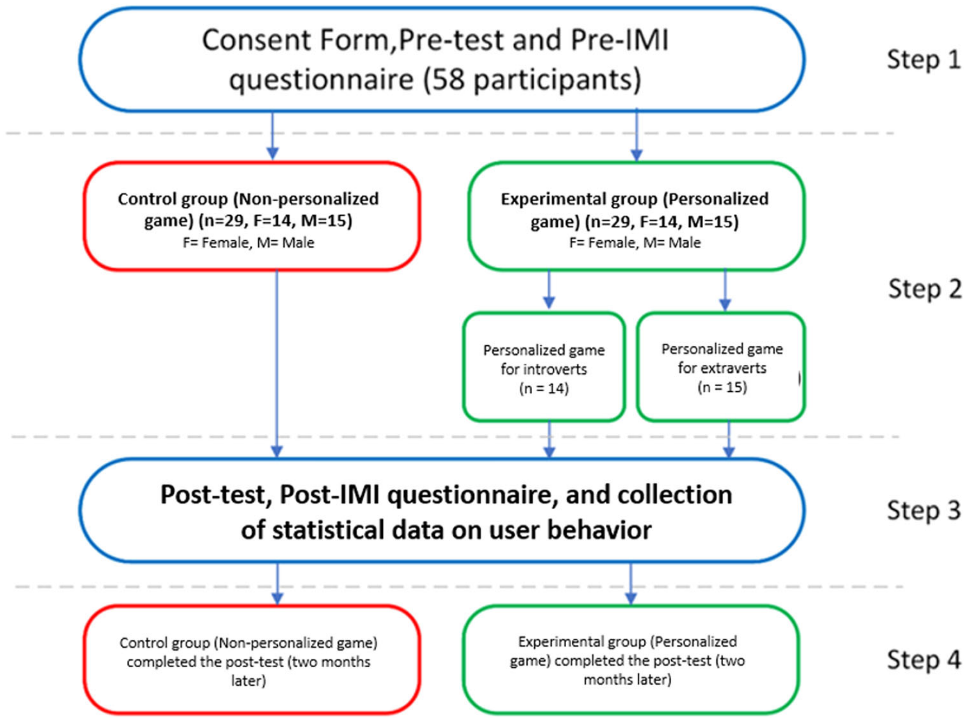

Fig. 6 Overview of research methodology.

out the post-IMI questionnaire to gauge their motivation levels linked to the serious game. Subsequently, we compared the pre-test and post-test scores to evaluate the impact of the serious game on knowledge acquisition and motivation, distinguishing between the Control and Experimental groups. The data collected from the questionnaires and pre/post-tests were analyzed using quantitative methods, utilizing the SPSS program. Finally, we extracted the statistical data from the web server to conduct an in-depth analysis of user behavior. This step was anticipated for participants to require approximately one hour to complete.

Step 4: Delayed post-test for knowledge retention. In the final step, we conducted a follow-up session with the same participants from both groups two months later, having them take the same post-test assessing knowledge acquisition in the designated room. Upon completing the examination, we provided all participants with a financial compensation of 200 baht. This compensation was allocated in the final step to ensure the participation of all individuals in the delayed post-test for knowledge retention, which was crucial for our data collection and for answering RQ2. Please note that even though we let participants take part in the survey by taking the questionnaire online, it still had to be done in the designated room order to avoid the use of any alternative tools for the questionnaire, which could have biased our data collection. This step was estimated to take approximately one hour to complete.

Data analysis. We collected and analyzed the results of the pre/post tests and the delayed post test, as well as the IMI questionnaire, to assess knowledge acquisition, retention and intrinsic motivation in information literacy, specifically in evaluating media sources. This analysis involved a comparison between the Control group (Non-personalized serious game) and the Experimental group (Personalized serious game). In the pre/post tests, we initially utilized the Shapiro-Wilk test to confirm the normal distribution of the data. Subsequently, we employed the parametric paired samples t-test to effectively evaluate knowledge acquisition, retention, and intrinsic motivation. A p-value of less than 0.05 was considered statistically significant. Additionally, we calculated the effect size using Cohen's d test. A small Cohen's d value of 0.2 suggests that the difference may not be very meaningful in the real world, while a large value, typically 0.8 or higher, indicates a practical and significant difference when comparing the two groups. Please note that, even though we let the participants do the survey questionnaire online, it still had to be done in the designated room to avoid the use of alternative tools for taking the questionnaire, which could have biased our data collection.

Results

The pre/post tests of knowledge acquisition and retention. The impact of the serious game on learning was assessed through the pre/post tests to address the first and second research questions. The results are presented in Tables 4, 5.

Table 4 presents the results focusing on knowledge acquisition and retention through pre/post tests. The pre-test score highlights participants' initial knowledge levels before any intervention (Control: M = 6.24, SD = 2.42; Experimental: M = 6.48, SD = 2.14). After completing the 2-weeks experiment, both groups showed significant improvement in knowledge acquisition, as indicated by their respective post-test scores (Control: M = 16.68, SD = 4.04; Experimental: M = 17.34, SD = 4.62). The results of the delayed post-test were M = 13.65, SD = 4.19 for the Control group and M = 14.37, SD = 4.22 for the Experimental group.

As shown in Table 5, the paired samples t-test conducted to compare the pre/post differences between the two groups was not significant ($p = 0.711$), suggesting that the difference in the effectiveness between the Non-personalized and Personalized serious game was not statistically significant; the small effect size (d = 0.296) also showed no practical significance. Comparison of the pre-test was not significant ($p = 0.619$) with a small, practically insignificant effect size (d = 0.272).

Table 4 Pre/post test results for knowledge acquisition and retention.

Group	*N*	Personality	Pre-test (SD)	Post-test (SD)	Post-test 2 month later (SD)
Control (Non-personalized game)	29 (F = 14, M = 15)	Introvert=15 Extravert=14	6.24 (2.42)	16.68 (4.04)	13.65 (4.19)
Experimental (Personalized game)	29 (F = 15, M = 14)	Introvert=14 Extravert=15	6.48 (2.14)	17.34 (4.62)	14.37 (4.22)

Table 5 Pre/post test analysis: t-test and Cohen's d.

Test	Group	Mean (Post-Pre)	Mean difference	Std. Deviation	Std. Error Mean	Sig. (2-tailed)	Cohen's D
Pre/Post (immediately)	Control Experimental	10.58 (4.98) 11.10 (5.10)	−0.517	7.443	1.382	0.711	0.296
Pre/delayed-Post (2 month later)	Control Experimental	7.55 (5.26) 8.13 (5.17)	−0.586	6.270	1.164	0.619	0.272

Table 6 Summary of responses to the IMI questionnaire, pre- and post-intervention for each group.

Group	IMI questionnaires	*N*	Pre (SD)	Post (SD)	Mean difference	*t*-value	Sig. (2-tailed)
Control (Non-personalized game)	Perceived Competence	29	3.20 (0.55)	3.27 (0.45)	−0.103	−1.361	0.184
	Interest		3.13 (0.51)	3.68 (0.47)	−0.551	−4.036	< 0.001***
	Effort		3.03 (0.62)	3.31 (0.47)	−0.275	−1.978	0.058
Experimental (Personalized game)	Perceived Competence	29	3.27 (0.45)	3.65 (0.66)	−0.379	−3.285	0.003**
	Interest		3.24 (0.43)	3.72 (0.52)	−0.482	−1.361	< 0.001***
	Effort		3.10 (0.48)	3.62 (0.49)	−0.517	−4.848	< 0.001***

**$p < 0.01$, ***$p < 0.001$

Regarding Tables 4 and 5, the data suggest that the use of personalized serious games among students at Chiang Mai University significantly enhanced knowledge acquisition and retention. However, this improvement was also observed in the control group, indicating that the utilization of a non-personalized serious game may have similarly enhanced knowledge acquisition and retention. Therefore, while both interventions led to improved outcomes, the personalized game did not demonstrate superiority over the non-personalized game in terms of knowledge acquisition and retention.

The pre/post tests of intrinsic motivation. The data on intrinsic motivation are summarized in Table 6, including the pre and post IMI questionnaires for both groups. For the Control group, the data showed a significant difference only in the dimension of Interest ($p < 0.001$). On the other hand, data for the Experimental group demonstrated a significant improvement in all dimensions, including Perceived competence ($p = 0.003$), Interest ($p < 0.001$), and Effort ($p < 0.001$).

Table 7 provides a comprehensive overview of the comparative analysis between the Control and Experimental groups using the IMI questionnaire. Accordingly, we first conducted the Shapiro-Wilk test to confirm the normal distribution of data. Subsequently, we utilized Welch's t-test for comparison, revealing a significant difference only in the dimension of Perceived Competence ($p = 0.035$). In contrast, the differences in the dimensions of Interest ($p = 0.714$) and Effort ($p = 0.175$) were not statistically significant.

Table 7 Results of the IMI questionnaire comparison using Welch's t-test between the Control and Experimental Groups.

IMI questionnaires	Group	*N*	Mean (Post-Pre)	Statistic[a]	Sig.(2-tailed)
Perceived Competence	Control Experimental	29 29	0.06 (0.45) 0.37 (0.62)	4.686	0.035*
Interest	Control Experimental	29 29	0.55 (0.73) 0.48 (0.68)	0.136	0.714
Effort	Control Experimental	29 29	0.27 (0.75) 0.51 (0.57)	1.890	0.175

*$p < 0.05$

Login data statistics (user behavior). The statistical analysis of login data for both groups aimed to discern the differences in user behavior between Personalized and Non-personalized serious games, providing insights into player engagement (Rapp, 2022). Fig. 7 presents the total playtime in minutes for both Control and Experimental groups over a 14-day period. Initially, both groups experienced a surge in playtime, peaking at slightly above 1000 min on the first day, followed by a sharp decline until the fourth day. Afterward, playtime stabilized around the 200 min mark for the remaining observation period in the Control group. In contrast, the Experimental group demonstrated a more gradual decline in playtime from the outset, reaching a minimum around day 7. Subsequently, there was a noticeable upward trend from day 8 to 14, indicating sustained player engagement over an extended period.

A more detailed comparison of the total playtime for extraverts in both groups is presented in Fig. 8 (Left), revealing that both the Experimental and Control groups exhibited consistently similar playtimes. Interestingly, for introverts, (Fig. 8, Right), playtime commenced at an apex close to 500 min on day one but swiftly dropped in the Control group, leveling off near 100 min by day 4 and continuing in this range with minor fluctuations for the remainder of the observation period. Conversely, the Experimental group demonstrated a more varied pattern, with playtime hovering around the 100 min mark for most days but a notable increase from day 12, culminating at nearly 250 min on day 14. Comparing this data to extraverted individuals, it's evident that while both personality types experienced a sharp decline in engagement with Non-personalized games, introverts displayed a more pronounced resurgence in playtime with Personalized games in the latter part of the observation window.

Table 8 provides a comprehensive summary of user behavior statistics for both groups, with participants categorized based on their personality traits as introverts and extraverts. In terms of total playtime, the Experimental group logged 3635 min, significantly surpassing the 2726 min recorded by the Control group. However, when examining the data by personality traits, a more nuanced pattern emerged. Notably, extraverted participants in both groups exhibited similar total playtime, with the Control group accumulating 1413 min and the Experimental group totaling 1527 min. The substantial contrast was apparent within the introvert category, where the Control group displayed a total playtime of 2108 min, considerably higher than the 1313 min logged by the Experimental group. This pattern is consistent when analyzing the average playtime per user and the number of logins. These data comparisons highlight that the personalized game had a more pronounced impact on increasing playtime and engagement, particularly among introverted participants, while for extraverted participants, the effect on total playtime was less pronounced.

In terms of measuring statistical significance, we compared the total playtime between introvert and extravert participants using Welch's t-test in both the Control and Experimental Groups, as shown in Table 9. The results align with the visual comparison of the means, indicating a significant difference in the personality trait of introversion ($p < 0.001$). On the other hand, the differences linked to extraversion ($p = 0.053$) were not found to be statistically significant.

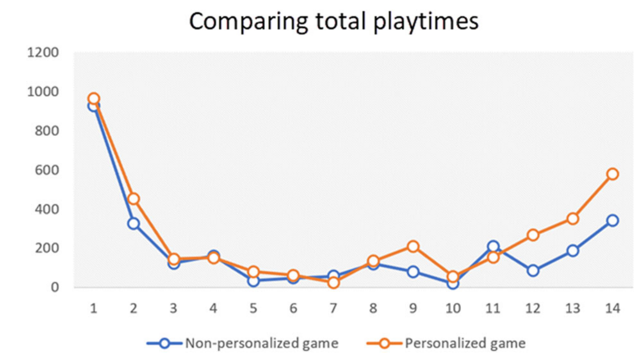

Fig. 7 Comparison of total playtimes between the Non-personalized and Personalized games.

Discussion and findings

Effects of the Personalized serious game. Concerning the effect of personalized serious games on knowledge acquisition (RQ1), the results of pre-post tests using a questionnaire related to information literacy and source evaluation revealed that the utilization of personalized serious gamesamong students at Chiang Mai University did not significantly enhance knowledge acquisition, as compared to the use of a non-personalized game. This outcome contradicts the findings of Tlili et al. (2019), who reported an improvement in knowledge acquisition by reducing the cognitive load during learning. Moreover, another study Smiderle et al. (2019) supported our results, indicating that both introverted and extraverted groups displayed enhanced performance in learning. However, the difference between the two groups was not statistically significant.

Regarding knowledge retention (RQ2), a post-test was conducted two months after the initial study to assess knowledge acquisition. The results revealed no significant difference between the two groups. Notably, there is a dearth of studies examining personalized games for introverts and extraverts concerning knowledge retention. Our findings indicate that the use of personalized serious games, incorporating suitable game elements, did not demonstrate a significant difference between introverted and extraverted individuals in terms of both knowledge acquisition and retention. This lack of distinction might be

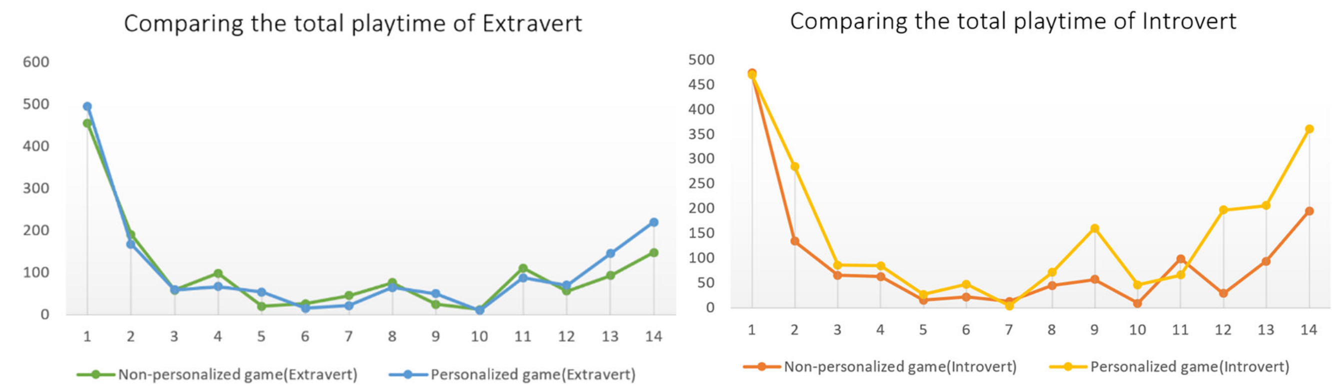

Fig. 8 Comparison of total playtimes between the two versions: Extravert (Left) and Introvert (Right).

Table 8 Summary of login data statistics.

Group	Participant	*N*	The average playtime per user (min)	Total time play(min)	Number of logins
Control (Non-personalized game)	Introvert	14	93.78	1313	57
	Extravert	15	94.20	1413	46
	Total	29	94.00	2726	103
Experimental (Personalized game)	Introvert	15	140.53	2108	77
	Extravert	14	109.07	1527	51
	Total	29	125.34	3635	128

Table 9 Comparison of playtimes between Introvert and Extravert participants using Welch's t-test in the Control and Experimental Groups.

Participant	Group	N	Mean (playtime)	Statistic	Sig.(2-tailed)
Introvert	Control	14	93.78 (15.82)	35.572	< 0.001***
	Experimental	15	140.53 (25.55)		
Extravert	Control	15	94.20 (20.17)	4.112	0.053
	Experimental	14	109.07 (19.31)		

***$p < 0.001$

attributed to the limited impact of individual game elements on knowledge acquisition (Landers et al., 2017).

Concerning intrinsic motivation (RQ3), the group using personalized games displayed significance across all aspects, including Perceived Competence, Interest, and Effort. In contrast, the Non-personalized game group exhibited significance only in the aspect of Interest. However, comparative analysis using Welch's t-test revealed a significant difference in Perceived Competence ($p = 0.035$). This suggests that the use of personalized serious games among students at Chiang Mai University significantly enhanced learning motivation, particularly in the dimension of Perceived Competence. Nonetheless, the observed improvement in Interest, regardless of game personalization, is consistent with previous studies indicating that educational games generally boost intrinsic motivation by providing fun and interest (Arayaphan et al., 2022; Leenaraj et al., 2023; Nieuwhof-Leppink et al., 2019). However, the divergent results in intrinsic motivation for both groups, specifically in terms of Perceived Competence and Effort, may be ascribed to the differential impacts of each game element on introverted and extraverted individuals. For instance, the Leaderboard game element (Jia et al., 2016) might induce pressure or anxiety in introverts due to concerns about their ranking, thereby leading to discomfort with attention and competitive stress. On the other hand, extraverts may experience positive effects from Leaderboards, as these features align with their inclination for social interaction, motivating them to compete and engage more actively with peers in the game.

The impact of player engagement in relation to extraversion and introversion (RQ4) was assessed through the analysis of login data statistics and usage, as well as statistical significance. The results revealed no statistically significant difference in engagement for extraverts across both groups, as measured by average and total playtime. However, a notable finding emerged for introverts, indicating a significant 37 percent increase in engagement, in terms of playtime for the Personalized version, compared to the Non-personalized version. This aligns with the results of Welch's t-test, which showed a very significant improvement among introverted individuals ($p < 0.001$). The observed difference in engagement among introverts within the context of personalized serious games might be attributed to their tendency to devote extended periods to exploring game mechanics, storylines, and intricate details. Introverts often engage deeply with games, appreciating their depth and complexity without external interruptions. In contrast, extraverts might experience more interruptions due to real-world activities, potentially affecting their engagement levels, even with a personalized version of the game.

Limitations and future research. The present study has limitations. It may face potential constraints in generalizability due to its focus on a specific demographic sample within a controlled educational setting. Factors such as varying access to technology and participants with devices could influence their experiences. Furthermore, the complex interactions between game elements and personality traits might pose challenges in identifying the most effective combinations, which could impact the outcomes and implications of the study. Addressing these aspects in future research endeavors will contribute to a more comprehensive understanding of the application and effectiveness of personalized serious games within diverse educational contexts.

Future research should aim to explore specific game elements or combinations that extend beyond introversion and extraversion. This exploration could enhance the parameters of personalization, optimizing the effectiveness of such games. Diversifying participant groups to encompass a broader demographic range would aid in understanding how various demographics respond to personalized serious games, considering factors such as age, cultural backgrounds, and educational diversity. Moreover, expanding the application of personalized serious games to different subjects or domains could provide valuable insights into the generalizability of this approach. Additionally, assessing the integration of emerging technologies like AI, virtual reality, or augmented reality into personalized serious games holds potential to enhance interactivity and improve learning outcomes.

Conclusions

This paper explores the development of serious games for information literacy and fake news, assessing the impact of serious games, personalized by incorporating suitable game elements tailored to students' personality traits, considering dimensions of introversion and extraversion. The study employs a quasi-experimental design to examine a serious game presenting two versions: one for the Non-personalized group and the other specifically tailored to suit the learning preferences of introverted and extraverted participants. The findings indicate the following:

- Personalization of serious games for educational purposes by means of suitable game elements for introverts and extraverts cannot effectively improve knowledge acquisition and retention compared to Non-personalized serious games.
- Personalized serious games can enhance intrinsic motivation, specifically in terms of perceived competence, which can be used as an interesting tool for educational environments and settings.
- Personalized serious games can significantly improve engagement, almost 37 percent in terms of average and total playtime for introverted people, but may not affect extraverted people.
- Designing personalized serious games by integrating suitable game elements for various personality traits might offer a cost-effective approach. However, careful consideration is essential regarding the selection of game elements, as their individual effectiveness may vary significantly.

Data availability

The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy and ethical reasons.

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Author contributions

Conceptualization, KT and PT; Methodology, KT; Software KT and PW; Validation SP and YT, Writing–original draft, KT and PT; Writing–review & editing, KT, KI, and PT. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This study was approved by the Chiang Mai University Research Ethics Committee (CMUREC No.64/092).

Informed consent

The informed consent was obtained from all participants.

Additional information

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Correspondence and requests for materials should be addressed to Kitti Puritat.